

# Nihal Machhi

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## SUMMARY

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Software Engineer curious about how systems work behind the scenes - from networks to infrastructure

## EDUCATION

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### Parul University

*Bachelor of Technology in Computer Science and Engineering(AI/ML) | CGPA – 7.07*

Vadodara, GJ

Sep. 2025 – May 2028

### Parul University

*Diploma in Information Technology | CGPA – 7.53*

Vadodara, GJ

Sep. 2022 – May 2025

## TECHNICAL SKILLS

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**Programming:** C, C++, Java, Python, JavaScript, TypeScript, SQL

**Full Stack Engineering:** React.js, Next.js, Node.js, Express.js, FastAPI, REST APIs

**AI & Machine Learning:** PyTorch, Hugging Face, LangChain, Pinecone, OpenCV, Gemini API

**Cloud & DevOps:** MongoDB, PostgreSQL, Firebase, Firestore, GCP, Supabase, Docker

**System Tools & Platforms:** Git/GitHub, Linux/Unix, Postman, Vercel, Streamlit

## PROJECTS

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### [DPI-Engine](#) | Python, FastAPI, Scapy, Jinja2, WebSockets, Uvicorn, Computer Networks

- Built real-time DPI engine using multi-threaded consistent-hash pipeline, hitting 10K+ packets/sec
- Implemented protocol fingerprinting via TLS SNI, HTTP Host, and DNS across 50+ live flows
- Designed producer-consumer pipeline with zero-copy buffers, validated by 34-test suite

### [MeshPay](#) | Python, FastAPI, Uvicorn, Cryptography, Pydantic, Pytest, HTTPX

- Designed offline UPI protocol using public-key crypto for internet-free P2P payments
- Built QR-based handshake with multi-device relay, blocking double-spending via signed receipts
- Developed async FastAPI endpoints, validated with Pytest/HTTPX suite covering edge cases

### [TrackitNow](#) | React, Vite, FastAPI, PostgreSQL, Supabase, Chart.js, Cloudinary, JWT

- Built full-stack habit tracker with 15+ tasks across 4 categories using React/FastAPI/PostgreSQL
- Engineered 12-month contribution heatmap using Chart.js with custom aggregation queries
- Secured 10+ REST endpoints with JWT auth and bcrypt, sanitizing against XSS/SQL injection

## EXPERIENCE

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### Placement coordinator Intern | Parul University

Aug 2024 – May 2025

- Coordinated 20+ recruitment drives by managing student-company data and placement workflows
- Collaborated with 15+ recruiters and placement teams, supporting 80+ student placements
- Maintained and updated placement databases, ensuring accurate student and recruiter records

## ACHIEVEMENTS

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- **Secured Runner-Up at KananHack 2026** (Kanan x Parul University) among 24 competing teams in a 24–48hr sprint
- **Built prototypes across 5 hackathons** — Google The Big Code, Odoo x Parul University, and HackerRank Orchestrate
- Solved 500+ DSA problems on LeetCode (arrays, graphs, trees, DP), reaching contest rating 1400+
- Applied Computer Networks and cryptography coursework to build DPI-Engine and MeshPay
- Coordinated placement operations for 80+ students as Placement Coordinator, managing 15+ recruitment drives
- Completed Anthropic's Claude API and Claude Code certifications, applying LLM tool-use and agentic workflows